INCH-POUND

A-A-52570
October 24, 1997

COMMERCIAL ITEM DESCRIPTION

TABLES, LINE-UP AND REGISTER, PRINTING, LITHOGRAPHIC

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for MIL-T-15672, for all federal agencies.

1. SCOPE This CID covers line-up and register tables used in the photolithographic process.

2. CLASSIFICATION

2.1. Type I. - 25.50 × 36.50 inches (647.7 mm × 927.1 mm) working area.

2.2. Type II. - 36.50 × 48.50 inches (927.1 mm × 1231.9 mm) working area.

3. SALIENT CHARACTERISTICS

3.1. Description. The table shall consist of a base frame with a shelf and lamp housing, a table top frame with a glass working surface, a horizontal and a vertical straightedge carriage, and precision scales.

3.2 Dimensions. Minimum dimensions of the glass area of tables shall be as shown in table I.

TABLE I. Table Dimensions.

Type I, inches (mm)	Type II, inches (mm)
28 × 39 (711.2 × 990.6)	39 × 51 (990.6 × 1295.4)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN: CETEC-TD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

AMSC N/A FSC 3610

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3.3. Base frame. The base frame shall be of sturdy construction and shall support the table and all of its components. A control switch shall be mounted on the base frame convenient for right-hand operation.

3.4. Tabletop frame. The tabletop frame shall form a plane and flat surface and shall provide a rigid support for the glass working surface and straightedges. The tabletop frame shall be adjustable for tilting up to 20° from horizontal.

3.5. Shelf. A shelf extending the entire length of the table shall be provided in the base frame.

3.6. Storage compartment. A minimum of one storage compartment or drawer shall be provided for the convenient storage of accessories and tools.

3.7. Power and illumination.

3.7.1. Electrical requirements. The table shall operate on 110-115 volt, 60 cycle, alternating current, single-phase, electrical service. All electrical equipment and wiring shall conform to NFPA No. 70, National Electrical Code.

3.7.2. Power cable. The table shall be provided with an insulated 3-wire power cable not less than 5 feet long, complete with a grounded plug conforming to NEMA WD 6 configuration 5-15P and UL-498 standards. The plug shall be for general purpose use.

3.7.3. Lamp housing. The lamp housing unit and reflector panels shall be of material and construction capable of providing rigid support and enclosure of the lamps and all electrical components. Access to facilitate removing and installing the lamps, and a means for dissipating heat from the lamps and the glass working surface shall be provided. The interior of the lamp housing shall have a durable heat-resistant finish providing even reflection to the glass area.

3.7.4. Illumination. The light source shall provide uniform illumination across the entire glass working surface with sufficient intensity for use with translucent and photographic materials.

3.8. Sheet stops and grippers. Stops and grippers shall be provided to accurately position and firmly hold the front edge of the sheet, negative, or lithographic plate. The stops shall not interfere in any way with the motion of the vertical straightedge assembly when they are in the secured position.

3.9. Side guide. A means shall be provided to accurately position the edge of the sheet, negative, or lithographic plate and to measure its position on the front fixed horizontal scale (see 3.18.1). The guide shall be pivoted so that it may be rotated into a position where it will not interfere with the motion of the straightedge assemblies.

3.10. Glass mounting frame. A rectangular frame shall be provided to mount the glass and diffuser in a firm, cushioned support. An adjustable support for the center of the glass and diffuser shall be provided.

3.11. Glass plate. The working surface shall be plate or float glass or other suitable material. When specified, extra glass plates shall be provided with each table (see 7.2).

3.11.1. Light diffusion. A means of diffusing the light over the entire working surface shall be provided. An opaline or milky white plastic sheet may be used. The light diffusion sheet shall be the same size as the glass plate.

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3.12. Straightedge carriages. Each table shall be equipped with one vertical and one horizontal straightedge carriage. Each carriage shall be provided with a straightedge, vernier scales, position lock, elevating mechanism, ruler guide track, and scales (see 3.16 and 3.18).

3.12.1. Straightedges. Each straightedge shall extend the full length or width of the glass working surface, and shall be movable across the entire working surface. Movement of the straightedges shall be parallel to their respective fixed scales. The spacings of the grid intersections shall be accurate and precise to within ± 0.0015 inch (.0381 mm). Extreme diagonals shall be equal to each other within 0.016 inch (0.406 mm) for type I tables and 0.022 inch (0.558 mm) for type II tables.

3.13. Vernier dial. Each carriage shall be equipped with a vernier dial calibrated to read directly the movement of its respective straightedge in .02 inch (.51 mm) intervals. A vernier adjusting screw, with a positive locking device, shall be provided for precise adjustment of the vernier dial.

3.14. Position lock. A means shall be provided for locking either straightedge carriage in any lateral position on the working surface. The ruling edge of the straightedge shall return to its original position on the table.

3.15. Elevating mechanism. Each straightedge carriage shall be equipped with a 3-position, lever-operated cam for raising and lowering the straightedge from: (1) surface contact with the sheet; (2) approximately .25-inch (6.35 mm) above the sheet; and (3) high elevation for off-table position. An adjustment shall be provided for variations in the thickness of paper, plates, or negatives.

3.16. Ruler guide track. A track shall be provided on each straightedge carriage to accurately guide the ruling device holder and self-inking ruling mechanism (see 3.16.1 and 3.16.3). A line drawn with either the ruling device holder or the self-inking mechanism shall be straight within ± 0.0015 inch (.0381 mm) per foot of length, and shall be parallel with the drawing edge of its respective straightedge.

3.16.1. Ruling device holder. A ruling device holder shall be provided for each straightedge carriage for ruling continuous parallel lines to within .09 inch (2.29 mm) of any edge of the working surface of the table. The holders shall be constructed for drawing lines on paper, tracing cloth, photographic film, and lithographic offset plates. The horizontal ruling device holder shall rule from left to right, and the vertical ruling device holder shall rule from back to front. Horizontal and vertical ruling device holders shall be provided with conspicuous, distinguishing marks.

3.16.2. Ruling pen. Each ruling device holder shall be provided with a ruling pen. The pen shall draw clean, unbroken uniform lines from 0.006 to 0.060 inch (.1524 mm to 1.524 mm) in width, without cutting into the drawing medium. The ruling pen shall have a 0.003 inch (.076 mm) and a 0.01 inch (.25 mm) stylus point.

3.16.3. Self-inking ruling mechanism. In addition to the ruling device holder specified in 3.16.1, a self-inking ruling mechanism shall be provided for each straightedge carriage. The self-inking mechanism shall rule continuous hairlines 0.003 inch (.076 mm) width to within 0.003 inch (.076 mm) of any edge of the working surface of the table. The self-inking ruling mechanism shall be provided with an ink pad, a roll assembly, and an ink wheel.

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3.17. Stops. Four movable and detachable stops, two for each straightedge carriage, shall be provided. Each stop shall be provided with an index for positioning the stop on the scale, and fine adjustment for limiting the stroke of either the ruling device holder or the self-inking ruling mechanism. The stops for the straightedge carriages shall be interchangeable.

3.18. Scales. All scales shall be of corrosion-resistant steel, shall be graduated in metric and inch divisions, and each inch graduation shall be numbered. In addition, the .12 inch (3.05 mm) graduations shall be marked with the proper figures expressed as the number of .03 inch (.76 mm). All graduations shall be machine divided, clear and distinct, and of a definite length for easy legibility.

3.18.1. Horizontal scales. One horizontal scale shall be fastened in front of the front edge of the glass working surface, and one on the horizontal straightedge carriage. Both horizontal scales shall have the zero graduation at the center of the glass working surface. Numbering shall increase to the right and left, and shall extend to the ends of the glass working surface.

3.18.2. Vertical scales. One vertical scale shall be fastened on the left-hand side (of the operator in normal working position) of the glass working surface, and one scale shall be fastened to the vertical straightedge carriage. Both vertical scales shall be numbered from front to back throughout the width of the glass working surface.

3.19. Lubrication. All moving parts shall be provided with suitable means of lubrication.

3.20. Treatment and painting. Unless otherwise specified (see 7.2), or unless the material is inherently corrosion resistant, the table shall be treated and painted for corrosion protection using the contractor's standard practice.

3.21. Identification of materials and finishes. The contractor shall identify and document the specific material and material finish or treatment of all components and sub-components, and shall make this information available upon request to the contracting officer or designated representative.

3.22. Fungus and moisture resistance. Materials used in the construction of this table shall resist mildew, rot, fungus and moisture encountered in tropical climates.

3.23. Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable (see 7.5.3).

4. REGULATORY REQUIREMENTS

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1. Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the contractor's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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5.2. Market acceptability. The product offered shall be of the latest model of the standard product of the supplier, and shall have been used in the field for at least one year by commercial organizations.

5.3. Examinations. Each line-up and register table examined shall be inspected for compliance with the requirements specified in 3.1 through 3.23. Any modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Nonconformance to any specified requirement, or the presence of one or more defects shall constitute cause for rejection.

6. PACKAGING

6.1. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1. Intended use. The tables are intended to be used in layout, lineup, registering, masking, and opaquing of lithographic negatives, plates, copy, prints, or other materials used in the photolithographic process.

7.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this CID.

b. When treatment and painting shall be other than the standard practice. (See 3.20).

c. When extra glass plates are required and number required (see 3.11).

d. Level of preservation and packing required (see 6.1).

7.3 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-XXXXX

AAXXXXX-2

Type II

7.4. Source of documents.

7.4.1. Copies of ANSI requirement documents are available from the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.

7.4.2. Copies of NFPA specifications are available from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.

7.4.3. Copies of (UL) Standard requirements documents are available from the Underwriters Laboratories, Inc., Publication Stock, 333 Pfingsten Road, Northbrook, Illinois 60062.

7.5. Definitions. For the purposes of this CID, the following definitions apply.

7.5.1. Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion.

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7.5.2. Fungus-inert. A fungus inert material is a material that is not a nutrient to fungi.

7.5.3. Interchangeable parts. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability. These parts are capable of being exchanged one for the other without alteration of the parts; of mating or adjoining parts, except for adjustment; and without selection for fit or performance.

7.5.4. Recovered materials. For the purpose of this requirement, recovered materials are those that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

7.5.5. Glass. For the purpose of this requirement, glass is any hard, more or less transparent substance.

7.6. Data requirements. The contracting officer should include, in the contract or order, any data requirements, such as operation and maintenance manuals, instructional materials, and technical publications to be furnished with each table.

7.7. Provisioning. The contracting officer should include in the contract or order, provisioning requirements for repair parts, necessary maintenance tools (including any special tools), and instructions regarding packaging and shipment of the tables and extra glass plates as required.

7.8. Subject term (key word) listing.

Compartment
Drawer
Glass top
Light
Lit
Photo lithographic

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